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Measurement-device-independent quantum key agreement based on entanglement swapping

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Abstract
Quantum key agreement (QKA) is an important cryptographic primitive that plays a pivotal role in private communications. However, in practical implementations of QKA, the flaws in participants' detectors may be exploited to compromise the security and fairness of the protocol. To address this issue, we propose a two-party measurement-device-independent QKA protocol, effectively eliminating all detector-side-channel loopholes. This protocol is based on quantum entanglement swapping and Bell-state measurements, making it feasible under current technological conditions. A thorough security analysis is conducted, demonstrating its ability to guarantee both security and fairness.

Keywords Quantum key agreement · Measurement-device-independent · Detector-side-channel attack · Quantum entanglement swapping · Bell-state measurement

1 Introduction

In recent years, securing group communication has garnered significant attention within the academic community. The approach to ensuring secure group communication revolves around maintaining a confidential key, known exclusively to all members of the group. This process of key generation is commonly referred to as group key establishment. There exist two principal methods to achieve this goal. The

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first is centralized key establishment, which involves a single party generating the group's secret key. This method is well-suited for two-party communications (such as client–server or peer-to-peer scenarios) as well as large multicast groups. However, in many collaborative group settings such as remote board meetings, teleconferences, whiteboard sessions, shared instrument control, secure and efficient data sharing, collaborative workspaces, cloud computing, and command-and-control systems becomes imperative. This entails employing distributed group key agreement techniques.

A key agreement protocol seeks to generate a shared conference key for multiple participants, ensuring the security of subsequent group communications in a manner where all participants exert influence on the outcome. Since its inception in the seminal paper by Diffie-Hellman [1], key agreement protocols have evolved into fundamental cryptographic primitives. However, classical key agreement protocols rely on public key cryptography, which hinges on computational complexity assumptions for security. The emergence of quantum computing has introduced new security threats to classical cryptosystems, leading to the development of quantum cryptography.

In 1984, Bennett and Brassard introduced the renowned BB84 quantum key distribution (QKD) protocol [2]. Since the inception of QKD, various branches of quantum cryptography have been developed, encompassing QKD [3, 4], quantum secure direct communication (QSDC) [5–8], quantum secret sharing (QSS) [9, 10], quantum private comparison (QPC) [11–13], quantum signature [14–16], quantum private query (QPQ) [17–24], quantum identity authentication (QIA) [25–27], quantum key agreement (QKA) [28–48] and anonymous quantum communication [49].

In 2004, Zhou et al. [28] introduced an innovative QKA protocol that utilizes quantum teleportation. However, Tsai and Hwang identified fairness concerns in Zhou et al.'s [29] protocol, highlighting its susceptibility to participant attacks, which enable one party to unilaterally determine the agreement key and distribute it to the other party without detection. Subsequently, a series of novel QKA protocols were proposed [30–48]. A secure QKA protocol should adhere to the following criteria: (C1) Correctness: At the protocol's conclusion, all participants should possess the correct agreement key provided they have executed the protocol honestly. (C2) Fairness: Each participant should wield an equivalent level of influence over the agreement key, rendering it impossible for any non-trivial subset of participants to unilaterally determine it. (C3) Security: The protocol should be designed such that external eavesdroppers cannot acquire information about the agreement key without triggering detection mechanisms.

While the security of quantum cryptographic protocols is theoretically guaranteed by the fundamental principles of quantum mechanics, their real-world implementations often encounter overlooked vulnerabilities, particularly in the imperfections of detector devices. Attackers can exploit these vulnerabilities to access agreement key information by employing quantum hacking techniques such as beam-splitting attacks, time-shift attacks, and blinding attacks [50–52]. Consequently, the burgeoning field of quantum hacking presents a significant threat to the practical security of quantum cryptographic protocols. To address the aforementioned challenges, Acín et al. [53] proposed the concept of Device-Independent QKD (DI-QKD) in 2006. However, the security of DI-QKD relies on loophole-free Bell tests between distant communication parties. Additionally, DI-QKD demands detectors with close-to-unit efficiency,

and its key generation rate is relatively low, rendering it impractical for real-world implementations.

Among all the concerns related to device security, vulnerabilities at the measurement device were historically considered the most challenging issue in the practical application of quantum cryptography. Various attacks, including phase remapping attacks, time-shift attacks, and SPD blinding attacks [50–52], have long stood as significant barriers to progress in this field. In response to these severe measurement device attacks and to address various shortcomings in DI protocols, researchers have introduced a specialized protocol known as the Measurement-Device-Independent (MDI) protocol. In 2012, Lo et al. [54] first introduced MDI-QKD. This protocol necessitates both communicating parties, Alice and Bob, to prepare the required single-qubit quantum states and transmit them to the untrusted third party, Charlie, for Bell state measurement. MDI-QKD remains immune to attacks targeting detectors, providing high key rates and communication distances. It effectively eliminates all detector side channels by assuming that legitimate participants possess trusted quantum state preparation devices. The field of MDI-QKD has witnessed numerous groundbreaking contributions. For example, the initial implementation of a free-space MDI-QKD protocol [55] was founded on a practical scheme from the literature [56], significantly surpassing the effective atmospheric thickness and paving the way for satellite-based MDI-QKD. Furthermore, various novel MDI-QKD protocols, such as twin-field QKD [57–61], mode-pairing QKD [62], and MDI-QKD network [63, 64], have been proposed.

However, there is a scarcity of research on MDI-QKA. Recently, Cai et al. [65] introduced a three-party MDI-QKA protocol. In this protocol, participant Charlie is tasked with measuring the Greenberger-Horne-Zeilinger (GHZ) state particle 'c' that he possesses, and his Z basis measurement results are employed for key generation. However, if an external eavesdropper were to target Charlie's measurement device and gain control over it, the measurement results could be pilfered by the eavesdropper, resulting in key leakage and posing a substantial threat to the security of the MDI-QKA protocol.

To eliminate all detector-side-channel vulnerabilities in QKA, this study presents an innovative two-party MDI-QKA protocol. This protocol leverages quantum entanglement swapping techniques, allowing participants to delegate measurement tasks to an untrusted third party, thereby eliminating the potential for detector-side-channel attacks. The proposed protocol delivers significantly enhanced practical security compared to existing two-party QKA protocols.

The subsequent sections of this paper follow this structure: Sect. 2 provides preliminaries, encompassing the characteristics of Bell states and the principles of quantum entanglement swapping. Section 3 introduces a two-party MDI-QKA protocol. Section 4 conducts a thorough analysis of the protocol, assessing correctness, fairness, and security. Finally, the paper concludes with a discussion and summary.

2 Preliminaries

Bell states are defined as follows: $|\phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$, $|\psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$. The principle of quantum entanglement swapping can be explained as follows. Suppose there are two EPR pairs which are in the Bell states $|\phi^{+}\rangle_{12} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $|\psi^{+}\rangle_{34} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$, respectively. The subscripts 1 and 2 represent the two qubits initially associated with the first Bell state, while the subscripts 3 and 4 represent the two qubits initially associated with the second Bell state. The state of the composite system can be rearranged by

$$\begin{aligned}
|\phi^{+}\rangle_{12} \otimes |\psi^{+}\rangle_{34} &= \frac{1}{2}(|00\rangle + |11\rangle)_{12} \otimes (|01\rangle + |10\rangle)_{34} \\
&= \frac{1}{2}(|0001\rangle + |1101\rangle + |0010\rangle + |1110\rangle)_{1234} \\
&= \frac{1}{2}(|01\rangle_{14}|00\rangle_{23} + |11\rangle_{14}|10\rangle_{23} + |00\rangle_{14}|01\rangle_{23} + |10\rangle_{14}|11\rangle_{23}) \\
&= \frac{1}{4}(|\psi^{+}\rangle + |\psi^{-}\rangle)_{14}(|\phi^{+}\rangle + |\phi^{-}\rangle)_{23} + (|\phi^{+}\rangle - |\phi^{-}\rangle)_{14}(|\psi^{+}\rangle - |\psi^{-}\rangle)_{23} \\
&\quad + (|\phi^{+}\rangle + |\phi^{-}\rangle)_{14}(|\psi^{+}\rangle + |\psi^{-}\rangle)_{23} + (|\psi^{+}\rangle - |\psi^{-}\rangle)_{14}(|\phi^{+}\rangle - |\phi^{-}\rangle)_{23} \\
&= \frac{1}{2}(|\psi^{+}\rangle_{14}|\phi^{+}\rangle_{23} + |\psi^{-}\rangle_{14}|\phi^{-}\rangle_{23} + |\phi^{+}\rangle_{14}|\psi^{+}\rangle_{23} - |\phi^{-}\rangle_{14}|\psi^{-}\rangle_{23}).
\end{aligned} \tag{1}$$

If we make Bell state measurements (BSM) on qubits 1 and 4, the state in Eq. (1) will collapse to one of the following four states $|\psi^{+}\rangle_{14}|\phi^{+}\rangle_{23}$, $|\psi^{-}\rangle_{14}|\phi^{-}\rangle_{23}$, $|\phi^{+}\rangle_{14}|\psi^{+}\rangle_{23}$, $|\phi^{-}\rangle_{14}|\psi^{-}\rangle_{23}$ with equal probability. Similarly, the results of quantum entanglement swapping for other combinations of initial Bell states are summarized in Table 1.

If we adopt the binary encoding "00" for $|\phi^{+}\rangle$, "01" for $|\phi^{-}\rangle$, "10" for $|\psi^{+}\rangle$, and "11" for $|\psi^{-}\rangle$, then the initial Bell states and the corresponding BSM outcomes will satisfy the following relationships:

$$O_{12} \oplus O_{34} = M_{14} \oplus M_{23}, \tag{2}$$

$$O_{12} \oplus M_{14} = O_{34} \oplus M_{23}, \tag{3}$$

$$O_{12} \oplus M_{23} = O_{34} \oplus M_{14}, \tag{4}$$

where O_{mn} and M_{ij} represent the binary representations of the initial Bell state of qubits m and n, and the BSM result on qubits i and$j$, respectively, and $\oplus$ denotes the bitwise XOR operation. The properties in Eqs. (2)–(4) will be used in the proposed MDI-QKA protocol.

Table 1 Entanglement swapping for different initial Bell states

	$\|\phi^+\rangle_{12}$	$\|\phi^-\rangle_{12}$	$\|\psi^+\rangle_{12}$	$\|\psi^-\rangle_{12}$
$\|\phi^+\rangle_{34}$	$\|\phi^+\rangle_{14}\|\phi^+\rangle_{23}$	$\|\phi^+\rangle_{14}\|\phi^-\rangle_{23}$	$\|\phi^+\rangle_{14}\|\psi^+\rangle_{23}$	$\|\phi^+\rangle_{14}\|\psi^-\rangle_{23}$
	$\|\phi^-\rangle_{14}\|\phi^-\rangle_{23}$	$\|\phi^-\rangle_{14}\|\phi^+\rangle_{23}$	$\|\phi^-\rangle_{14}\|\psi^-\rangle_{23}$	$\|\phi^-\rangle_{14}\|\psi^+\rangle_{23}$
	$\|\psi^+\rangle_{14}\|\psi^+\rangle_{23}$	$\|\psi^+\rangle_{14}\|\psi^-\rangle_{23}$	$\|\psi^+\rangle_{14}\|\phi^+\rangle_{23}$	$\|\psi^+\rangle_{14}\|\phi^-\rangle_{23}$
	$\|\psi^-\rangle_{14}\|\psi^-\rangle_{23}$	$\|\psi^-\rangle_{14}\|\psi^+\rangle_{23}$	$\|\psi^-\rangle_{14}\|\phi^-\rangle_{23}$	$\|\psi^-\rangle_{14}\|\phi^+\rangle_{23}$
$\|\phi^-\rangle_{34}$	$\|\phi^+\rangle_{14}\|\phi^-\rangle_{23}$	$\|\phi^+\rangle_{14}\|\phi^+\rangle_{23}$	$\|\phi^+\rangle_{14}\|\psi^-\rangle_{23}$	$\|\phi^+\rangle_{14}\|\psi^+\rangle_{23}$
	$\|\phi^-\rangle_{14}\|\phi^+\rangle_{23}$	$\|\phi^-\rangle_{14}\|\phi^-\rangle_{23}$	$\|\phi^-\rangle_{14}\|\psi^+\rangle_{23}$	$\|\phi^-\rangle_{14}\|\psi^-\rangle_{23}$
	$\|\psi^+\rangle_{14}\|\psi^-\rangle_{23}$	$\|\psi^+\rangle_{14}\|\psi^+\rangle_{23}$	$\|\psi^+\rangle_{14}\|\phi^-\rangle_{23}$	$\|\psi^+\rangle_{14}\|\phi^+\rangle_{23}$
	$\|\psi^-\rangle_{14}\|\psi^+\rangle_{23}$	$\|\psi^-\rangle_{14}\|\psi^-\rangle_{23}$	$\|\psi^-\rangle_{14}\|\phi^+\rangle_{23}$	$\|\psi^-\rangle_{14}\|\phi^-\rangle_{23}$
$\|\psi^+\rangle_{34}$	$\|\phi^+\rangle_{14}\|\psi^+\rangle_{23}$	$\|\phi^+\rangle_{14}\|\psi^-\rangle_{23}$	$\|\phi^+\rangle_{14}\|\phi^+\rangle_{23}$	$\|\phi^+\rangle_{14}\|\phi^-\rangle_{23}$
	$\|\phi^-\rangle_{14}\|\psi^-\rangle_{23}$	$\|\phi^-\rangle_{14}\|\psi^+\rangle_{23}$	$\|\phi^-\rangle_{14}\|\phi^-\rangle_{23}$	$\|\phi^-\rangle_{14}\|\phi^+\rangle_{23}$
	$\|\psi^+\rangle_{14}\|\phi^+\rangle_{23}$	$\|\psi^+\rangle_{14}\|\phi^-\rangle_{23}$	$\|\psi^+\rangle_{14}\|\psi^+\rangle_{23}$	$\|\psi^+\rangle_{14}\|\psi^-\rangle_{23}$
	$\|\psi^-\rangle_{14}\|\phi^-\rangle_{23}$	$\|\psi^-\rangle_{14}\|\phi^+\rangle_{23}$	$\|\psi^-\rangle_{14}\|\psi^-\rangle_{23}$	$\|\psi^-\rangle_{14}\|\psi^+\rangle_{23}$
$\|\psi^-\rangle_{34}$	$\|\phi^+\rangle_{14}\|\psi^-\rangle_{23}$	$\|\phi^+\rangle_{14}\|\psi^+\rangle_{23}$	$\|\phi^+\rangle_{14}\|\phi^-\rangle_{23}$	$\|\phi^+\rangle_{14}\|\phi^+\rangle_{23}$
	$\|\phi^-\rangle_{14}\|\psi^+\rangle_{23}$	$\|\phi^-\rangle_{14}\|\psi^-\rangle_{23}$	$\|\phi^-\rangle_{14}\|\phi^+\rangle_{23}$	$\|\phi^-\rangle_{14}\|\phi^-\rangle_{23}$
	$\|\psi^+\rangle_{14}\|\phi^-\rangle_{23}$	$\|\psi^+\rangle_{14}\|\phi^+\rangle_{23}$	$\|\psi^+\rangle_{14}\|\psi^-\rangle_{23}$	$\|\psi^+\rangle_{14}\|\psi^+\rangle_{23}$
	$\|\psi^-\rangle_{14}\|\phi^+\rangle_{23}$	$\|\psi^-\rangle_{14}\|\phi^-\rangle_{23}$	$\|\psi^-\rangle_{14}\|\psi^+\rangle_{23}$	$\|\psi^-\rangle_{14}\|\psi^-\rangle_{23}$

3 Proposed MDI-QKA protocol

We consider a scenario where two participants Alice and Bob, seek to establish an agreement key K in a secure and fair manner over a quantum channel. First, Alice and Bob prepare their respective EPR pairs (a_1,a_2) and (b_3,b_4), where each EPR pair is randomly in one of the four Bell states $\{|\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle\}$. Then, Alice and Bob transmit particles a_2 and b_3 to the untrusted third party Charlie, respectively. Charlie performs BSM on particles a_2 and b_3 and announces the measurement outcome M_{23}. Following that, Alice and Bob send the remaining particles a_1 and b_4 to Charlie for BSM, and Charlie announces the measurement outcome M_{14}. Finally, Alice can compute the key $K_A = O_{12} \oplus M_{14}$, and Bob can compute the key $K_B = O_{34} \oplus M_{23}$. It is evident that $K_A = K_B$. So, the agreement key can be chosen as $K = K_A = K_B$. The two-party MDI-QKA protocol is illustrated in Fig. 1.

The flowchart of the protocol is illustrated in Fig. 2.

Next, let's describe the details of the two-party MDI-QKA protocol as follows:

(1) Alice and Bob have pre-agreed that the four Bell states $\{|\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle\}$ can encode two bits of classical information according to the rules: $|\phi^+\rangle$: "00", $|\phi^-\rangle$: "01", $|\psi^+\rangle$: "10", and $|\psi^-\rangle$: "11".

(2) Alice (Bob) prepares a large number of EPR photon pairs each of which is randomly in one of the four Bell states $\{|\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle\}$. Alice (Bob) selects

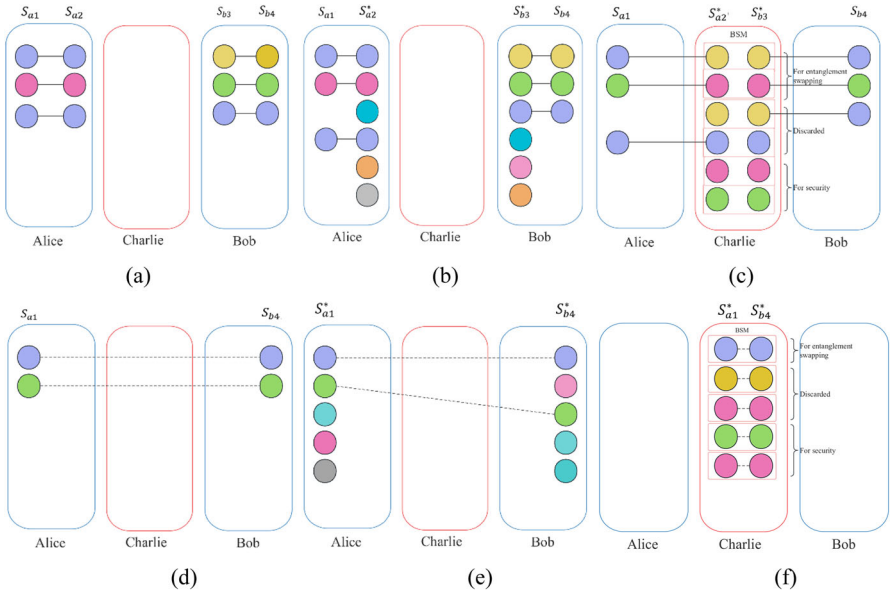


Fig. 1 (Color online) The schematic diagram of a two-party MDI-QKA protocol based on entanglement swapping. In **a**–**c**, the particles connected by solid lines represent the EPR pairs used for potential key generation. The colored single particles denote single photons employed for eavesdropping detection. In **d**–**f**, particles connected by dotted lines depict EPR pairs generated through entanglement exchange. Different-colored particle pairs connected by solid lines represent four distinct Bell states, namely, $|\phi^+\rangle$, $|\phi^-\rangle$, $|\psi^+\rangle$, $|\psi^-\rangle$. The red rectangular boxes symbolize BSM, and the blue boxes denote single-particle measurements in the Z or X basis

a photon of each EPR pair to form the S_{a1} (S_{b3}}) sequence, and the other photon from each photon pair forms the sequence S_{a2} (S_{b4}). Alice (Bob) stores the S_{a1} (S_{b4}) sequence in the quantum memory devices. Meanwhile, Alice (Bob) prepares a set of decoy particles, denoted as $\widetilde{R}_{a_2}(\widetilde{R}_{b_3})$, which are enough for eavesdropping check. Each decoy particle is randomly prepared in either the Z-basis ($|0\rangle$, $|1\rangle$) or the X-basis ($|+\rangle = 1/\sqrt{2}(|0\rangle + |1\rangle)$, $|-\rangle = 1/\sqrt{2}(|0\rangle - |1\rangle)$). Subsequently, Alice (Bob) randomly inserts the decoy photons into the $S_{a2}(S_{b3})$ sequence and generates the new sequence denoted as $S^*_{a2}(S^*_{b3})$. Finally, the sequence $S^*_{a2}(S^*_{b3})$ is sent to Charlie.

(3) After the BSM on each received photon pair, Charlie announces the BSM results. Then, Alice and Bob announce the positions and quantum states of the decoy photons in a random order. According to the positions of decoy photons, there are three different scenarios. In the first scenario, the two photons for the BSM come from two EPR pairs. In this scenario, the corresponding remaining photons of these two EPR pairs form a new Bell state by the entanglement swapping. In the second scenario, for a BSM, one photon is from the EPR pair and the other one is a decoy photon. In this scenario, Alice and Bob should discard the BSM result and the remaining photon of the EPR pair. In the third scenario, the photons for a BSM are both decoy photons. When the two photons are prepared

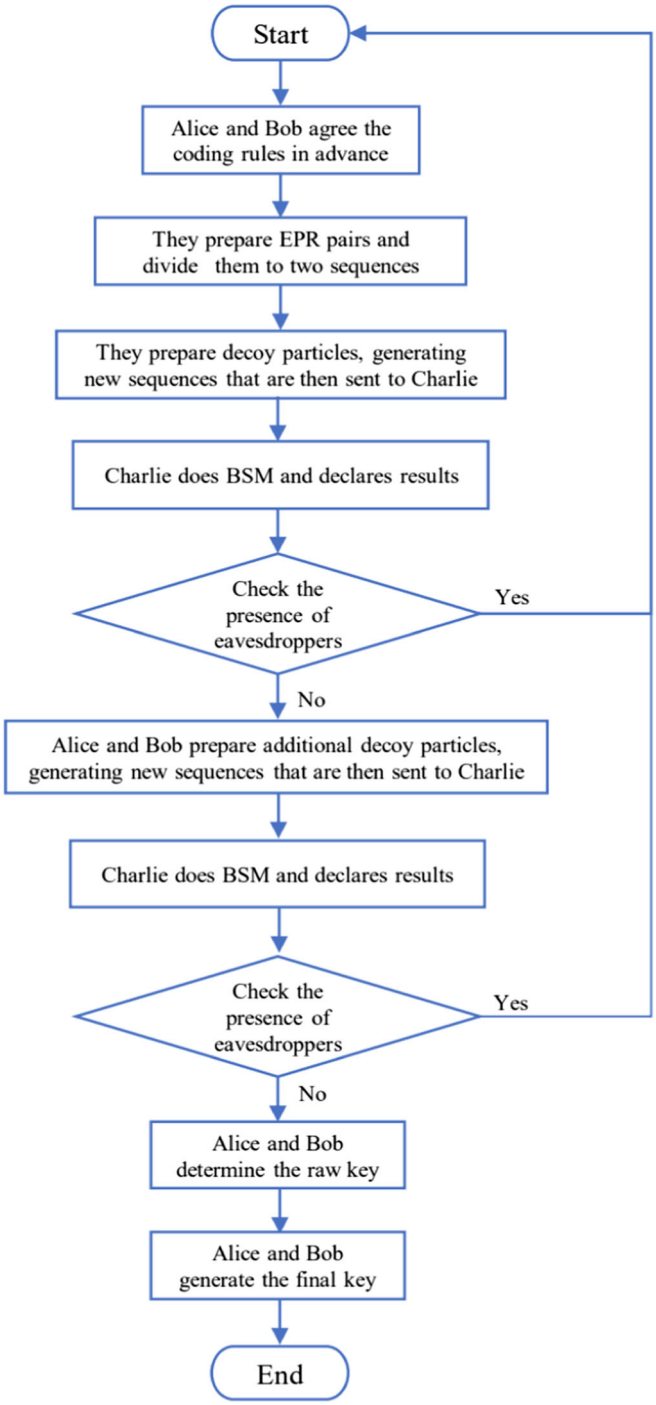


Fig. 2 The flowchart of the MDI-QKA protocol

Table 2 The relationships between the BSM result and the decoy states

Quantum states		The probability of possible BSM results			
Alice	Bob	$\|\phi^+\rangle$	$\|\phi^-\rangle$	$\|\psi^+\rangle$	$\|\psi^-\rangle$
$\|0\rangle$	$\|0\rangle$	0.5	0.5	0	0
$\|0\rangle$	$\|1\rangle$	0	0	0.5	0.5
$\|1\rangle$	$\|0\rangle$	0	0	0.5	0.5
$\|1\rangle$	$\|1\rangle$	0.5	0.5	0	0
$\|+\rangle$	$\|+\rangle$	0.5	0	0.5	0
$\|+\rangle$	$\|-\rangle$	0	0.5	0	0.5
$\|-\rangle$	$\|+\rangle$	0	0.5	0	0.5
$\|-\rangle$	$\|-\rangle$	0.5	0	0.5	0

with the same basis, the BSM result can be used for checking security. The relationships between the BSM result and the decoy states are shown in Table 2. For example, the states of Alice and Bob are $|0\rangle_{\widetilde{R}_{a_2}}|0\rangle_{\widetilde{R}_{b_3}}$ which can be written as $|0\rangle_{\widetilde{R}_{a_2}}|0\rangle_{\widetilde{R}_{b_3}} = \frac{1}{\sqrt{2}}(|\phi^+\rangle_{\widetilde{R}_{a_2}\widetilde{R}_{b_3}} + |\phi^-\rangle_{\widetilde{R}_{a_2}\widetilde{R}_{b_3}})$. In this way, both $|\phi^\pm\rangle_{\widetilde{R}_{a_2}\widetilde{R}_{b_3}}$ correspond to the correct results. If the BSM result is $|\psi^\pm\rangle_{\widetilde{R}_{a_2}\widetilde{R}_{b_3}}$, the parties can find an error. After the security checking, if the error rate exceeds the threshold, Alice and Bob have to discard the communication and restart step (2). Otherwise, they confirm the security of the first photon transmission round and go on to the next step.

(4) Alice (Bob) prepares the other decoy photon sequence, denoted as $\widetilde{R}_{a_1}(\widetilde{R}_{b_4})$, and randomly inserts $\widetilde{R}_{a_1}(\widetilde{R}_{b_4})$ into the $S_{a1}(S_{b4})$ sequence to generate the new sequence denoted as $S^*_{a1}(S^*_{b4})$. Finally, the sequence $S^*_{a1}(S^*_{b4})$ is sent to Charlie.

(5) After the BSM on each received photon pair, Charlie announces the BSM results. Then, Alice and Bob announce the positions and quantum states of the decoy photons in a random order. If one photon is from the EPR pair and the other one is a decoy photon, Alice and Bob should discard the BSM result. If the photons for a BSM are both decoy photons and the two photons are prepared with the same basis, the BSM result can be used for the second round of security checking. The security checking rule is the same as that in step 3. If the error rate exceeds the tolerable threshold, the second round of photon transmission is insecure and thus the BSM results should be discarded. Otherwise, the security of the second round of photon transmission can be guaranteed.

Finally, after the two rounds of photon transmission, the remained BSM results corresponding to the sequences S_{a2} and S_{b3} are denoted as $M^1_{a2,b3}$, $M^2_{a2,b3}$, ..., $M^N_{a2,b3}$, and the remained BSM results corresponding to the sequences S_{a1} and S_{b4} are denoted as $M^1_{a1,b4}$, $M^2_{a1,b4}$, ..., $M^N_{a1,b4}$. According to Table 1, Charlie's BSM results and her initial Bell states, Alice can determine the raw agreement key $K\prime = O_{a1,a2} \oplus M_{a1,b4}$. And according to Table 1, Charlie's BSM results and his initial

Bell states, Bob can determine the raw agreement key $K\prime = O_{b3,b4} \oplus M_{a2,b3}$. Alice and Bob keep the generated raw agreement key $K\prime$.

(6) Alice and Bob apply error correction and privacy amplification on $K\prime$ to generate the final agreement key K.

4 Analysis of correctness, fairness, security and efficiency

4.1 Correctness

Theorem 1 Assuming that Alice and Bob are honest, they can achieve a consensus on the agreement key K.

Proof If Alice and Bob faithfully follow the protocol, it is evident that they can collaboratively generate the raw agreement key $K\prime$. This is achieved through Charlie's successful implementation of BSM on the received photon pairs in the two rounds of photon transmission. Assume that the two received photon pairs in the two rounds of photon transmission for the BSM come from two EPR pairs, for example, $(P_i(a_1),\ P_i(b_4))$ and $(P_i(a_2),\ P_i(b_3))$, which results in quantum entanglement swapping and the measurement results are presented in Table 1. In Table 1, O_{mn} represents the binary encoding of the initial states of EPR particles m and n, while M_{ij} denotes the binary representation of the BSM results for EPR particles i and j. Notably, the relationships hold, i.e., $O^i_{a1,a2} \oplus M^i_{a2,b3} = O^i_{b3,b4} \oplus M^i_{a1,b4}$ and $O^i_{a1,a2} \oplus M^i_{a1,b4} = O^i_{b3,b4} \oplus M^i_{a2,b3}$, for $i = 1, 2, \ldots, N$. Therefore, according to Table 1, Charlie's BSM results and her initial Bell states, Alice can determine the raw agreement key $K\prime = O_{a1,a2} \oplus M_{a1,b4}$. And according to Table 1, Charlie's BSM results and his initial Bell states, Bob can determine the raw agreement key $K\prime = O_{b3,b4} \oplus M_{a2,b3}$. Based on this, Alice and Bob can perform error correction and privacy amplification on $K_A\prime$ and $K_B\prime$ to obtain the final agreement key K, respectively.

4.2 Fairness

Theorem 2 There is no subset of participants that has the capability to determine the agreement key K alone.

Proof The proposed MDI-QKA protocol is symmetric, with identical roles for Alice and Bob. According to Theorem 1, it can be deduced that if either of Alice and Bob intends to determine the final agreement key K, they must first generate the raw agreement key $K\prime$, which, in practical terms, is unattainable. The raw agreement key $K\prime$ is derived from the information of the initial Bell states of EPR pairs held by Alice and Bob as well as Charlie's BSM results. Here, we assume Alice is dishonest and her purpose is to determine the raw agreement key $K\prime$ alone, and succeed in cheating Bob to accept this fake key. The same conclusion can be obtained for dishonest Bob. The first strategy is the intercept-resend attack, i.e., Alice tries to intercept the photons sent by Bob and replace them with fake photons. However, the decoy particles are

randomly distributed in the sequence. Moreover, the density operators of the decoy photons and the reduced density operator of one photon of the Bell state are same, i.e., the decoy photons and the EPR particles are in the maximally mixed state $\rho = \frac{1}{2}($ $|0\rangle\langle 0| + |1\rangle\langle 1|)$. Thus, Alice cannot obtain any information about only from Bob's traveling particle. Since the agreement key is generated by $K\prime = O_{b3,b4} \oplus M_{a2,b3}$ which depends on Bob's initial Bell state, Alice cannot determine the raw agreement key $K\prime$ alone without the knowledge of Bob's initial Bell state even if she performs a BSM attack on her and Bob's traveling particles.

4.3 Security

Security means that no external eavesdropper can obtain the information about the agreement key without being detected. In our protocol, the security of all the photon transmission processes is guaranteed by the two rounds of security checking. Suppose Eve is an external eavesdropper who attempts to eavesdrop on the agreement key. Since the decoy photons and the EPR particles are in the maximally mixed state $\rho = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|)$, Eve cannot discriminate the decoy photons and the EPR particles. Thus, Eve cannot obtain any information only from Alice's and Bob's traveling particles. To see this in a sufficient way, we will consider two possible cases, in which Eve may execute two common attacks: intercept-resend attack and collective attack. For the intercept-resend attack, it is unavoidable for Eve to intercept some decoy photons. As Eve does not know the bases of the decoy photons, she may choose wrong measurement bases. Obviously, the probability that Eve selects a wrong measurement basis is approximately 50%. In this way, after she sends the photons to Charlie, Charlie may obtain a BSM result which may mismatch Alice's and Bob's decoy photon states and increase the error rate of the security checking. As a result, the intercept-resend attack can be detected by Alice and Bob.

Another more general attack is the collective attack. The general idea of this attack is that Eve performs a collective attack on both Alice-Charlie and Bob-Charlie channels. Concretely, first, Eve prepares two ancillary particles, i.e., e_1 and e_2, in the initial states, $|0\rangle_{e_1}$ and $|0\rangle_{e_2}$. In step 2, when the sequence $S^*_{a2}\left(S^*_{b3}\right)$ is sent to Charlie, Eve entangles the first ancilla e_1 with the jth particle in the sequence S^*_{a2}, and then sends the jth particle to Charlie, and she also entangles the second ancilla e_2 with the jth particle in the sequence S^*_{b3}, and then sends the jth particle to Charlie. Eve tries to obtain the information about Alice's and Bob's initial Bell states by measuring two ancillary particles. In the following, we will show that Eve cannot obtain the information about Alice's and Bob's initial Bell states under the condition that no error occurs.

Here, we can write the most general operations, E_1 and E_2, which Eve could apply to Alice's and Bob's particles and two ancillary particles, as follows:

$$E_1 : |0\rangle_a|0\rangle_{e_1} \rightarrow |0\rangle_a|\varepsilon_{00}\rangle_{e_1} + |1\rangle_a|\varepsilon_{01}\rangle_{e_1},\ |1\rangle_a|0\rangle_{e_1} \rightarrow |0\rangle_a|\varepsilon_{10}\rangle_{e_1} + |1\rangle_a|\varepsilon_{11}\rangle_{e_1}, \tag{5}$$

$$E_2 : |0\rangle_b|0\rangle_{e_2} \rightarrow |0\rangle_b|\tau_{00}\rangle_{e_2} + |1\rangle_b|\tau_{01}\rangle_{e_2},\ |1\rangle_b|0\rangle_{e_2} \rightarrow |0\rangle_b|\tau_{10}\rangle_{e_2} + |1\rangle_b|\tau_{11}\rangle_{e_2}, \tag{6}$$

where the states $|\varepsilon_{ij}\rangle$ and $|\tau_{ij}\rangle$ are pure ancilla states uniquely determined by E_1 and E_2. The following conditions can be derived from the unitary features of these two operations:

$$\begin{aligned}\langle\varepsilon_{00}|\varepsilon_{10}\rangle + \langle\varepsilon_{01}|\varepsilon_{11}\rangle = 0,\ \langle\tau_{00}|\tau_{10}\rangle + \langle\tau_{01}|\tau_{11}\rangle = 0,\\ \langle\varepsilon_{00}|\varepsilon_{00}\rangle + \langle\varepsilon_{01}|\varepsilon_{01}\rangle = \langle\varepsilon_{10}|\varepsilon_{10}\rangle + \langle\varepsilon_{11}|\varepsilon_{11}\rangle = 1,\\ \langle\tau_{00}|\tau_{00}\rangle + \langle\tau_{01}|\tau_{01}\rangle = \langle\tau_{10}|\tau_{10}\rangle + \langle\tau_{11}|\tau_{11}\rangle = 1.\end{aligned} \tag{7}$$

To be useful for the security check, the states of Alice's and Bob's decoy particles must be one of the following combinations $\{|0\rangle_a|0\rangle_b,|0\rangle_a|1\rangle_b,|1\rangle_a|0\rangle_b,|1\rangle_a|1\rangle_b,$ $|+\rangle_a|+\rangle_b,|+\rangle_a|-\rangle_b,|-\rangle_a|+\rangle_b,|-\rangle_a|-\rangle_b\}$. For simplicity and without loss of generality, we assume that the combined state of Alice's and Bob's decoy particles is $|0\rangle_a|0\rangle_b$.

In the collective attack, Alice's and Bob's particles and the two ancillary particles are in an initial state $|\alpha^0\rangle_{abe_1e_2} = |0\rangle_a|0\rangle_b|0\rangle_{e_1}|0\rangle_{e_2}$. After Eve's two unitary interactions between particles a and e_1 and between b and e_2, the whole system including Alice's and Bob's particles and the two ancillary particles is in the state

$$\begin{aligned}|\alpha^1\rangle_{abe_1e_2} &= E_1|0\rangle_a|0\rangle_{e_1}E_2|0\rangle_b|0\rangle_{e_2}\\ &= \frac{1}{\sqrt{2}}\big[|\phi^+\rangle_{ab}\big(|\varepsilon_{00}\rangle_{e_1}|\tau_{00}\rangle_{e_2} + |\varepsilon_{01}\rangle_{e_1}|\tau_{01}\rangle_{e_2}\big)\\ &\quad + |\phi^-\rangle_{ab}\big(|\varepsilon_{00}\rangle_{e_1}|\tau_{00}\rangle_{e_2} - |\varepsilon_{01}\rangle_{e_1}|\tau_{01}\rangle_{e_2}\big)\\ &\quad + |\psi^+\rangle_{ab}\big(|\varepsilon_{00}\rangle_{e_1}|\tau_{01}\rangle_{e_2} + |\varepsilon_{01}\rangle_{e_1}|\tau_{00}\rangle_{e_2}\big)\\ &\quad + |\psi^-\rangle_{ab}\big(|\varepsilon_{00}\rangle_{e_1}|\tau_{01}\rangle_{e_2} - |\varepsilon_{01}\rangle_{e_1}|\tau_{00}\rangle_{e_2}\big)\big].\end{aligned} \tag{8}$$

In the case without eavesdropping, the composite state of Alice's and Bob's particles is $|0\rangle_a|0\rangle_b$ which can be written as $|0\rangle_a|0\rangle_b = \frac{1}{\sqrt{2}}(|\phi^+\rangle_{ab}+|\phi^-\rangle_{ab})$. In this way, both $|\phi^\pm\rangle_{ab}$ correspond to the correct results. If Charlie's BSM result is $|\psi^\pm\rangle_{ab}$, Alice and Bob will find an error. In the eavesdropping check process, in order for there to be no error, the state $|\alpha^1\rangle_{abe_1e_2}$ should satisfy the following conditions:

$$\langle\psi^+|\alpha^1\rangle_{abe_1e_2} = 0,\ \langle\psi^-|\alpha^1\rangle_{abe_1e_2} = 0. \tag{9}$$

From Eqs. (8) and (9), the following equation can be derived:

$$|\varepsilon_{00}\rangle|\tau_{01}\rangle = |\varepsilon_{01}\rangle|\tau_{00}\rangle = \overrightarrow{0}, \tag{10}$$

where $\overrightarrow{0}$ denotes a zero vector.

For other state combinations $\{|0\rangle_a|1\rangle_b,|1\rangle_a|0\rangle_b,|1\rangle_a|1\rangle_b,|+\rangle_a|+\rangle_b,|+\rangle_a|-\rangle_b,$ $|-\rangle_a|+\rangle_b,\ |-\rangle_a|-\rangle_b\}$, we obtain the following equations:

$$\begin{aligned}|\varepsilon_{00}\rangle|\tau_{10}\rangle = |\varepsilon_{01}\rangle|\tau_{11}\rangle = \overrightarrow{0},\ |\varepsilon_{10}\rangle|\tau_{00}\rangle = |\varepsilon_{11}\rangle|\tau_{01}\rangle\\ = \overrightarrow{0},\ |\varepsilon_{10}\rangle|\tau_{11}\rangle = |\varepsilon_{11}\rangle|\tau_{10}\rangle = \overrightarrow{0}.\end{aligned} \tag{11}$$

Based on Eq. (11), we obtain the following constraints:

$$|\varepsilon_{01}\rangle = |\varepsilon_{10}\rangle = |\tau_{01}\rangle = |\tau_{10}\rangle = \overrightarrow{0}. \tag{12}$$

The constraint in Eq. (12) implies that Eve cannot obtain any information under the condition that no errors are introduced. Hence, the proposed protocol is secure against the collective attack.

On the other hand, we need to consider the attack from Charlie. Charlie's dishonest behavior means that he announces wrong BSM results deliberately. This behavior may cause Alice and Bob to obtain wrong agreement keys, so that it can disturb the key agreement. However, as Charlie does not know the position of the decoy photons before announcing the BSM result, it is unavoidable that Charlie announces wrong BSM results for a decoy photon pair. As a result, Charlie's dishonest behavior will increase the error rate in both security checking processes, and thus can be checked out by Alice and Bob. If the error rate exceeds the tolerable threshold, the key agreement will be terminated and the generated agreement keys will be discarded. Meanwhile, although two rounds of BSM results are announced, no one except Alice and Bob can obtain the generated agreement keys, for only Alice and Bob know their respective initial Bell states.

4.4 Efficiency

Efficiency of a quantum cryptographic protocol is defined as [66]

$$\eta = \frac{c}{q+b}, \tag{13}$$

where the parameter c represents the length of the final secret key, q denotes the total number of qubits utilized, and b indicates the classical bits exchanged for decoding the secret key. In the context of two-party MDI-QKA protocols, disregarding the resources consumed by security checks, for two pair of Bell states generated by Alice and Bob, as well as Charlie's four-bit BSM results, a 2-bit key is generated. Thus, the efficiency of our two-party MDI-QKA protocol is

$$\eta = \frac{2}{4+4} = 25\%. \tag{14}$$

In Table 3, we provide a comparison with existing two-party QKA protocols [30, 31, 35, 36, 41, 43, 44, 47, 48].

From Table 3, although the efficiency of our two-party MDI-QKA protocol is lower than that of other existing ones [31, 35, 36, 41], it can remove all the detector-side channel loopholes thereby having a higher practical security.

Table 3 Performance comparison between our protocol and the protocols [30, 31, 35, 36, 41, 43, 44, 47, 48]

Scheme	QR	NQO	QE (%)
Ref. [30]	Single particles	SQM + SQUO	16.67
Ref. [31]	Cluster states	FQOM + SQUO	26.67
Ref. [35]	Two-particle states	SQM + SQUO	33
Ref. [36]	EPR pairs	BSM	33.33
Ref. [41]	Cluster states	FQOM + SQUO	30.77
Ref. [43]	Single particles	SQM	16.67
Ref. [44]	Single particles	SQM + SQUO	25
Ref. [47]	Single particles	BSM + SQUO	16.67
Ref. [48]	EPR pairs	SQM	25
Ours	Single particles, EPR pairs	BSM	25

QR Quantum resource, *NQO* Necessary quantum operation, *QE* Quantum efficiency, *SQUO* Single-qubit unitary operation, *SQM* Single-qubit measurement, *FQOM* Four-qubit orthogonal measurement, *BSM* Bell basis measurement

5 Discussion and conclusion

Our work represents more than a mere extension of existing research. Currently, QSDC/DSQC has been proposed for QKA protocols. However, a majority of the current QKA protocols remain susceptible to detector-side-channel vulnerabilities. Remarkably few studies have ventured into MDI-QKA, with only the protocols in [65, 67] tackling this domain. Nonetheless, even among the existing MDI-QKA studies [65, 67], substantial practical flaws persist. First and foremost, the protocol in [65] does not qualify as a genuine MDI-QKA protocol, as it remains susceptible to security attacks. In this protocol, Charlie, a participant in the key agreement process, is tasked with measuring the GHZ state particle 'c' using his measurement device to generate the key based on the measurement result. However, Charlie's measurement device may be eavesdropped upon or even controlled by an external attacker, compromising the entire protocol's security. For an MDI-QKA protocol claiming to withstand all side-channel vulnerabilities, the existence of such a vulnerability is catastrophic. Recently, Liu et al. [67] presented a multiparty MDI-QKA protocol. In their protocol, the post-selected GHZ states are used to generate a secure agreement key between legitimate participants. However, GHZ-state measurements are required and this protocol is probabilistic because only two of 2^N N-particle GHZ-state measurement results can be identified by using the linear optical elements. In contrast, our two-party MDI-QKA protocol is deterministic. Since multi-party MDI-QKA is more

desirable for practical quantum networks, it is natural to study the deterministic multiparty MDI-QKA protocol based on entanglement swapping. In fact, the two-party MDI-QKA can be used as a subprotocol for the construction of multiparty MDI-QKA. For concreteness, assume that there be M participants who intend to engage in key negotiation, divided into m pairs (where $M = 2\,m$): $\{Alice_1, Bob_1\}$, $\{Alice_2, Bob_2\}$, …, $\{Alice_m, Bob_m\}$. They encode the four Bell states according to the rules: $|\phi^+\rangle$: "00", $|\phi^-\rangle$: "01", $|\psi^+\rangle$: "10", and $|\psi^-\rangle$: "11". Each pair $\{Alice_i, Bob_i\}$ run the two-party MDI-QKA protocol and generate a two-party agreement key K_i, for $i = 1$, 2, …, m. For each $Alice_i(Bob_i)$, they transmit their respective K_i to $Alice_j(Bob_j)$ using the MDI-QSDC method [68] where $j \in \{1, 2, \cdots, m\}\backslash\{i\}$. Subsequently, the raw key $K\prime$ is obtained as:

$$K\prime = K_1 \oplus K_2 \oplus ... \oplus K_m. \tag{15}$$

Finally, they utilize error correction and privacy amplification techniques on $K\prime$ to generate the final agreement key K. However, it is easily seen that this generalization is inefficient due to so many rounds of invoking two-party MDI-QKA and MDI-QSDC protocols. So, we will study the efficient multiparty MDI-QKA in the future work.

To summarize, we have presented an innovative two-party MDI-QKA protocol based on entanglement swapping. Significantly, this protocol guarantees that no subset of participants can independently ascertain the final shared key. Moreover, we also show that the proposed protocol is secure against the intercept-resend attack and collective attack of external attackers, and it is also secure against participant attacks, thereby ensuring both fairness and security. Furthermore, the generation of Bell states and single photons is achievable with current technologies. In addition, our protocols need complete Bell state analysis. Though Bell state measurement has not been generally accomplished [69], it was experimentally realized based on some certain techniques [70–73]. Furthermore, the realizations of entanglement swapping have been proposed [74]. Therefore, our protocols are within the reach of current technology. Finally, we hope that the MDI technique should be taken into account in the design of practical quantum cryptographic protocols [75–92].

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Data availability No data was used for the research described in the article.

Declarations

Conflict of interests We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript.

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